Document 04: EDI 837 & 835 ANSI X12 Generation

CodePerfect Auditor — Healthcare Claims Interoperability Engine

Project: CodePerfect Auditor | **Version:** 1.0 | **Date:** 31-03-2026 **Submitted To:** Virtusa Hackathon | **Institution:** Saveetha Engineering College

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Overview

The most technically demanding integration in the entire CodePerfect Auditor platform is the generation of **ANSI ASC X12 EDI** healthcare transaction sets. EDI (Electronic Data Interchange) is the legally mandated machine-readable format that US federal law (HIPAA Transaction Standards — 45 CFR Part 162) requires all insurance claims to be transmitted in.

CodePerfect implements two complementary EDI transaction sets:

Transaction	Standard	Direction	Purpose
EDI 837P	`005010X222A1`	Hospital → Payer	Submit a claim for payment
EDI 835	`005010X221A1`	Payer → Hospital	Acknowledge payment or denial

The raw output looks horrifying to humans — but it is perfectly structured for machine parsing by payer clearinghouses. This is exactly what makes it so impressive to judges: most hackathon teams never go near legacy EDI. We built a full, spec-compliant generator.

Part 1: EDI 837P — Claim Submission — `services/edi_837_builder.py`

X12 Format Fundamentals

```
# ── Constants ──────────────────────────────────────────────────────────────
_SEGMENT_TERMINATOR = "~"     # Every segment ends with tilde
_ELEMENT_SEP        = "*"     # Fields within a segment are separated by asterisk
_SUBELEMENT_SEP     = ":"     # Sub-fields (composite elements) use colon
_SENDER_ID          = "INTGRNX01"
_VERSION            = "005010X222A1"   # HIPAA-mandated version for institutional
claims
```

```
_TRANSACTION_TYPE    = "837"
_FUNCTIONAL_ID       = "HC"             # "HC" = Health Care (used in GS segment)
```

A raw EDI file for a single claim looks like this:

```
ISA*00*          *00*          *ZZ*INTGRNX01      *ZZ*GLBLHLTH
*260331*0030*^*00501*000000001*0*T*:~
GS*HC*INTGRNX01*GLBLHLTH*20260331*0030*1*X*005010X222A1~
ST*837*0001*005010X222A1~
BHT*0019*00*9B3862E1E4E742*20260331*0030*CH~
NM1*41*2*ORG 1 HOSPITAL*****46*INTGRNX01~
NM1*40*2*GLOBAL HEALTH INSURANCE*****46*GLBLHLTH~
HL*1**20*1~
PRV*BI*PXC*208D00000X~
NM1*85*2*ORG 1 HOSPITAL*****XX*1234567890~
HL*2*1*22*0~
SBR*P********11~
NM1*IL*1*UNKNOWN****MI*9B3862E1E4E742~
CLM*9B3862E1E4E742*541.00***11:B:1*Y*A*Y*I~
DTP*434*RD8*20260331-20260331~
HI*ABK:DC11.0XA8KL9*ABF:DC11.0XA0077~
LX*1~
SV1*HC:22045*541.00*UN*1***1~
DTP*472*D8*20260331~
SE*17*0001~
GE*1*1~
IEA*1*000000001~
```

Helper Functions — The Building Blocks

```python
def _seg(*elements: str) -> str:
    """Join elements with * and terminate every segment with the ~ tilde."""
    return _ELEMENT_SEP.join(str(e) for e in elements) + _SEGMENT_TERMINATOR

def _money(value) -> str:
    """Convert any numeric value or None to a safe 2-decimal-place string.
    EDI spec requires monetary values EXACTLY as '0.00' — never '0' or '0.0'.
    An incorrect format causes automated payer rejection without explanation."""
    try:
        return f"{float(value):.2f}"
    except (TypeError, ValueError):
        return "0.00"

def _alpha(s: Optional[str], max_len: int = 35) -> str:
    """Return uppercase, alphanumeric-safe string, truncated to max_len.
    EDI strictly prohibits special characters — commas, apostrophes, or
    hyphens in a hospital name will corrupt the segment delimiter structure."""
    if not s:
```

```
        return ""
    cleaned = re.sub(r"[^A-Za-z0-9 ]", "", s).strip().upper()
    return cleaned[:max_len]

def _edi_date(iso_str: Optional[str]) -> str:
    """Convert ISO 8601 date-time (YYYY-MM-DD) to EDI format YYYYMMDD.
    EDI uses 8-digit condensed dates — never dashes or slashes."""
    if not iso_str:
        return datetime.now(timezone.utc).strftime("%Y%m%d")
    clean = iso_str.strip()[:10]
    return clean.replace("-", "")   # "2026-03-31" → "20260331"
```

Explanation

These helper functions are the critical foundation of X12 compliance. Every character in an EDI file matters. The `_alpha()` function strips special characters because an apostrophe in "St. Mary's Hospital" would be interpreted as a sub-element delimiter by the clearinghouse parser, corrupting the entire claim transaction. The `_money()` function enforces exactly 2 decimal places — a value of `541` instead of `541.00` triggers an automatic `999 Rejection Acknowledgement` from the payer.

Patient Demographics — Extracting from FHIR

```
def _extract_patient(fhir_claim: dict) -> dict:
    """
    Pull structured patient data from the contained Patient resource inside the
    FHIR Claim. Returns a flat dict with keys: first, last, full, dob_edi,
sex_code.

    The FHIR Claim embeds a Patient resource in its 'contained' array.
    This is FHIR R4 standard — we never store raw patient data separately.
    """
    contained: list = fhir_claim.get("contained") or []
    patient_resource = next(
        (r for r in contained if r.get("resourceType") == "Patient"), {}
    )

    # ── Name ──────────────────────────────────────────────────────────────
    names: list = patient_resource.get("name") or []
    first = last = full = ""
    if names:
        name_entry = names[0]
        full  = _alpha(name_entry.get("text"), 40)
        last  = _alpha(name_entry.get("family"), 35)
```

```
        given = name_entry.get("given") or []
        first = _alpha(given[0] if given else "", 25)

    # ── Date of Birth ──────────────────────────────────────────────────
    raw_dob  = patient_resource.get("birthDate")       # YYYY-MM-DD from FHIR
    dob_edi  = _edi_date(raw_dob) if raw_dob else None  # None = DMG segment
omitted

    # ── Sex ────────────────────────────────────────────────────────────
    fhir_gender = patient_resource.get("gender", "")   # "male" | "female" |
"other"
    sex_map     = {"male": "M", "female": "F"}
    sex_code    = sex_map.get(fhir_gender.lower(), "U") # U = Unknown

    return {"first": first, "last": last, "full": full,
            "dob_edi": dob_edi, "sex_code": sex_code}
```

Explanation

The `DMG` segment (Demographics) that carries `dob_edi` is intentionally omitted if no Date of Birth is extracted from the clinical chart. This is correct HIPAA-compliant behaviour — CMS auditors flag claims that emit placeholder DOBs (e.g., `19000101`) as fraudulent data fabrication. The system is designed to "emit nothing" rather than "emit fake data," which is a non-trivial compliance engineering decision.

The Main EDI 837 Builder — Full Segment Walk-Through

```
def build_edi_837(
    *,
    fhir_claim: dict,
    org_name: str,
    payer_name: str,
    claim_db_id: str,
    total_billed_amount: float,
    service_date: Optional[str] = None,
) -> str:
    """
    Generate a complete ANSI ASC X12 837P EDI transaction set.
    Returns a newline-separated string — each line is one EDI segment ending with ~
    """
    now          = datetime.now(timezone.utc)
    isa_date     = now.strftime("%y%m%d")   # ISA uses 6-digit YYMMDD (legacy
requirement)
    gs_date      = now.strftime("%Y%m%d")   # GS uses 8-digit YYYYMMDD (modern)
```

```
    # Use first 20 chars of claim UUID as the submitter claim control number.
    # This is how the payer clearinghouse matches the 835 response back to this
claim.
    claim_control = claim_db_id.replace("-", "")[:20]

    patient = _extract_patient(fhir_claim)
    segments: list[str] = []

    def add(*elements: str):
        segments.append(_seg(*elements))
```

Segment 1: ISA — Interchange Control Header

```
    # ISA: Interchange Control Header — the outermost envelope of the EDI
transaction.
    # Every EDI file starts with exactly one ISA and ends with one IEA.
    add(
        "ISA", "00", "          ", "00", "          ",
        "ZZ", _pad_right(sender_id, 15),    # ISA06: sender ID (padded to 15 chars)
        "ZZ", _pad_right(receiver_id, 15),  # ISA08: receiver ID (padded to 15
chars)
        isa_date, isa_time,
        "^",            # ISA11: repetition separator (005010 uses ^, unlike
004010)
        "00501",        # ISA12: interchange version number
        _pad_left(interchange_ctrl, 9),     # ISA13: interchange control number (9
digits)
        "0",            # ISA14: 0 = no acknowledgement requested
        "T",            # ISA15: T = Test, P = Production
        _SUBELEMENT_SEP,  # ISA16: sub-element separator is the : colon
    )
```

Why ISA15 = "T"? We emit `T` (Test mode) in the codebase to prevent accidentally routing real hospital claims to payer clearinghouses during development. This would be changed to `P` (Production) during live deployment — a single character change with enormous compliance implications.

Segment 2: Claim Information (CLM)

```
    # CLM: The core financial segment of the entire EDI file.
    # This is the segment the payer's adjudication engine reads first.
    add(
        "CLM",
        claim_control,                              # CLM01: submitter claim ID
(our UUID)
        _money(total_billed_amount),                # CLM02: TOTAL BILLED
($541.00)
```

```
        "",                                             # CLM03: not required
        "",                                             # CLM04: not required
        "11" + _SUBELEMENT_SEP + "B" + _SUBELEMENT_SEP + "1",
        # CLM05: "11:B:1" = Place of Service (11=Office) : Facility Type (B) :
Claim Frequency (1=Original)
        "Y",            # CLM06: Y = provider accepts assignment (mandatory)
        "A",            # CLM07: A = benefits assigned to provider
        "Y",            # CLM08: Y = signed release of information on file
        "I",            # CLM09: I = patient signature on file
    )
```

Diagnosis Codes — The HI Segment

```
    # HI: Health Information segment — carries ALL ICD diagnosis codes.
    # Format: HI*ABK:PRIMARY_CODE*ABF:SECONDARY_CODE*ABF:ADDITIONAL_CODE
    #   ABK = principal ICD diagnosis (first code on the claim)
    #   ABF = additional ICD diagnosis (comorbidities, secondary codes)
    if fhir_diagnoses:
        hi_elements = ["HI"]
        for idx, dx in enumerate(fhir_diagnoses):
            codings = dx.get("diagnosisCodeableConcept", {}).get("coding") or []
            code     = codings[0].get("code", "").strip()
            qualifier = "ABK" if idx == 0 else "ABF"  # First = principal, rest =
secondary
            hi_elements.append(qualifier + _SUBELEMENT_SEP + code)
        add(*hi_elements)
        # Example output: HI*ABK:DC11.0XA8KL9*ABF:DC11.0XA0077~
```

CPT Service Lines — The LX / SV1 Segments

```
    # For each CPT procedure code resolved by Node 3 (CPT Resolver):
    for idx, item in enumerate(fhir_items, start=1):
        cpt_code   = cpt_codings[0].get("code", "").strip()  # e.g. "22045"
        unit_price = item.get("unitPrice", {}).get("value") or 0.0
        quantity   = item.get("quantity", {}).get("value") or 1

        add("LX", str(idx))    # LX: Line Number — separates each service line
        add(
            "SV1",
            "HC" + _SUBELEMENT_SEP + cpt_code,  # SV101: "HC:22045" (Health Care +
CPT code)
            _money(unit_price),                 # SV102: charge for THIS procedure
            "UN",                               # SV103: UN = Unit
            str(int(quantity)),                 # SV104: number of service units
            "", "",                             # SV105-106: not required in 837P
            "1",                                # SV107: diagnosis pointer → links
to HI position 1
```

```
        )
        add("DTP", "472", "D8", edi_service_date)  # Service date for this line
item
```

Explanation

The `SV107` field "diagnosis code pointer" (`1`) is only one character but it is legally critical. It tells the payer adjudicator that this CPT procedure was clinically justified by the FIRST diagnosis code in the HI segment. If a claim has 3 diagnosis codes but a procedure only links to pointer `2`, the adjudicator knows to pay using the second code's DRG weight. Incorrect pointers cause multi-million-dollar claim denials in live hospital billing systems.

Part 2: EDI 835 — Payment/Remittance Advice — `services/edi_835_builder.py`

The EDI 835 is the **payer's response**. After the payer adjudicates the submitted 837, they return an 835 telling the hospital exactly how much they are paying and why.

Status Code Mapping

```
# X12 CLM02 status codes — the code that tells the hospital the outcome
_CLM_STATUS: dict[str, str] = {
    "PAID":           "1",   # 1 = Processed as Primary (full payment)
    "PARTIALLY_PAID": "2",   # 2 = Processed as Secondary (partial pay)
    "DENIED":         "4",   # 4 = Denied
}

# CAS Group Codes — categorize WHY an adjustment was made
_CAS_GROUP = {
    "PAID":           "CO",  # CO = Contractual Obligation (write-off due to
contract)
    "PARTIALLY_PAID": "CO",
    "DENIED":         "OA",  # OA = Other Adjustments
}

# CARC = Claim Adjustment Reason Codes — precise reason for every dollar difference
_CARC_CODE = {
```

```
    "PAID":           "45",  # 45 = Exceeds contracted maximum (standard
contractual write-off)
    "PARTIALLY_PAID": "45",
    "DENIED":         "96",  # 96 = Non-covered charge(s)
}
```

The BPR and CLP Segments — Financial Heart of the 835

```
def build_edi_835(
    *, claim_id, claim_status, total_billed, total_paid,
    org_name, payer_name, patient_name=None, service_date=None,
    fhir_claim=None, denial_reason=None
) -> str:

    total_adjustments = total_billed - total_paid  # The contractual write-off
amount

    # ── BPR: Beginning of Payment ──────────────────────────────────────
    add(
        "BPR",
        "I",                      # BPR01: I = Information (no EFT), C = Check, A =
ACH
        _money(total_paid),       # BPR02: Total amount being paid (0.00 for
denied)
        "C",                      # BPR03: C = Credit
        "NON",                    # BPR04: NON = non-electronic payment
        "", "", "", "", "", "",    # BPR05-10: bank routing (not applicable for
info-only)
        payment_date,             # BPR16: payment effective date
    )

    # ── CLP: Claim-Level Payment Summary ──────────────────────────────
    # This is the single most important segment in the 835.
    # The hospital's revenue management system reads CLP01 to match this 835
    # back to the original 837 claim using the claim_control UUID.
    add(
        "CLP",
        claim_control,            # CLP01: matches the CLM01 from the original
837
        clm_status_code,          # CLP02: 1=paid, 2=partial, 4=denied
        _money(total_billed),     # CLP03: what the hospital asked for
        _money(total_paid),       # CLP04: what the payer is actually paying
        _money(max(patient_responsibility, 0.0)),  # CLP05: patient deductible/co-
pay
        "11",                     # CLP06: 11 = Professional claim type
        "INT" + claim_control[:10], # CLP07: payer's internal claim reference
number
    )
```

Claim Adjustment Segment (CAS) — Explaining Every Dollar Not Paid

```
    # ── CAS: Claim Adjustment Segment ──────────────────────────────────────
    # The CAS segment is required whenever the payer pays LESS than the billed
amount.
    # This is how the payer legally justifies the write-off to the hospital.
    # Without a CAS, the hospital's billing system cannot properly post the
payment.
    if abs(total_adjustments) > 0.001:
        add("CAS", cas_group, carc_code, _money(total_adjustments))
        # Example for a DENIED claim:
        # CAS*OA*96*541.00~
        # Translation: "Other Adjustment, Reason Code 96 (Non-covered charge),
Amount $541.00"
```

Service-Level Payment Detail

```
    # ── SVC: Service Line Payment ──────────────────────────────────────────
    # One SVC segment per CPT code — tells the hospital exactly how much
    # was paid for EACH individual procedure (versus the aggregate total).
    for idx, item in enumerate(fhir_items, start=1):
        cpt_code   = cpt_codings[0].get("code", "")
        unit_price = float(item.get("unitPrice", {}).get("value") or 0.0)

        # Proportionate paid amount: if total was $541 and we're paying $400,
        # each line gets paid proportionally based on its share of the total
billed.
        if total_billed > 0:
            line_paid = round(unit_price * (total_paid / total_billed), 2)
        else:
            line_paid = 0.0

        add(
            "SVC",
            "HC" + _SUBELEMENT_SEP + cpt_code,  # SVC01: "HC:22045"
            _money(unit_price),                  # SVC02: amount billed for this
line
            _money(line_paid),                   # SVC03: amount paid for this line
        )
        if abs(unit_price - line_paid) > 0.001:
            add("CAS", cas_group, carc_code, _money(unit_price - line_paid))
```

Explanation

The line-level `SVC→CAS` pairing is what allows a hospital's accounting department to post payments accurately to individual patient ledgers. A claim with 5 CPT codes generates 5 SVC segments, each individually reconciled. This is the difference between a toy prototype and a real-world RCM integration.

Part 3: The Claims API Integration — `routes/claims.py`

The EDI builders are invoked from the `/api/v1/claims` route when the payer adjudicates or the hospital exports a claim.

```
class ClaimSubmissionRequest(BaseModel):
    """Pydantic model — validates every field at the API boundary before any DB
write."""
    session_id:          str
    organization_id:     str
    payer_id:            str
    patient_name:        Optional[str] = None
    patient_dob:         Optional[str] = None
    patient_sex:         Optional[str] = None
    total_billed_amount: float
    claim_data:          dict    # Contains icd_codes, cpt_codes, financial_summary
    submission_notes:    Optional[str] = None

@router.post("/submit")
async def submit_claim(req: ClaimSubmissionRequest):
    """
    Submits a finalized coding session as a Claim to the Payer.
    After FastAPI validates the Pydantic model, the endpoint:
    1. Checks for duplicate submissions (prevents double-billing)
    2. Runs the Payer Policy Gate (auto-approve evaluation)
    3. Builds the FHIR Claim Proposal
    4. Writes the claim row to PostgreSQL with status=SUBMITTED
    """
    supabase = get_supabase()

    # Duplicate guard: one coding session can produce at most one claim.
    existing = supabase.table("claims").select("id").eq("session_id",
req.session_id).execute()
    if existing.data:
        raise HTTPException(status_code=400,
                            detail="A claim has already been submitted for this
session.")
```

Payer Policy Auto-Gate

```
    # Run the Payer Policy Gate before writing any database record.
    # This evaluates whether the payer's configured auto-approve rules
    # allow this claim to skip manual adjudication entirely.
    gate_report = run_payer_policy_gate(
        claim_data=req.claim_data,
        payer_policy=payer_policy or {},
        org_settings=org_settings,
    )

    # Embed the gate report in the claim_data JSONB payload.
    # The payer adjudicator UI reads this to understand WHY a claim
    # was auto-approved or flagged for manual review.
    req.claim_data["payer_gate_report"] = gate_report
```

Explanation

The **Payer Policy Gate** (`payer_policy_gate.py`) is a deterministic rule engine that evaluates each submitted claim against the payer's configured thresholds:

- `auto_approve_confidence_min` — minimum AI confidence score required
- `auto_approve_max_risk` — maximum risk score allowed for auto-approval
- `auto_approve_requires_patient_dob` — DOB must be present for auto-approval
- `accepted_icd_versions` — payer may reject ICD-10 or ICD-11 claims

If all conditions pass, the claim status is automatically set to `PAID` without requiring a human adjudicator — this is the core automation value proposition of CodePerfect Auditor for insurance companies.

End-to-End EDI Flow

```
Hospital Coder submits claim via UI
         |
         ▼
POST /api/v1/claims/submit
  → Pydantic validation (ClaimSubmissionRequest)
  → Duplicate guard check
  → Payer Policy Gate evaluation
```

```
  → FHIR Claim Proposal generated (build_fhir_claim_proposal)
  → Claim inserted into PostgreSQL (status = SUBMITTED)
        │
        ▼
Payer Views Claim in /payer/inbox
  → Runs adjudication (PAID / DENIED / PARTIALLY_PAID)
  → POST /api/v1/claims/{id}/adjudicate
        │
        ▼
EDI 835 Generated (build_edi_835)
  → Downloadable by Hospital for payment posting
        │
        ▼
EDI 837 Generated (build_edi_837) on demand
  → Hospital downloads for external clearinghouse submission
  → Machine-readable ANSI X12 format — directly loadable into
    Epic, Cerner, or any clearinghouse (Availity / Change Healthcare)
```
